

Assignment of Chapter One

A. Fill-in-the-blank Questions:

1. Digital circuits can only process _____ signals, but they cannot process _____ signals.
2. The machine code consists of _____, _____, and _____ such three types of code.
3. A binary number, 1001 1001, its decimal number is _____, its hex number is _____.
4. A binary number, 111 0101, its decimal number is _____, its octal number is _____.
5. A binary number, 0.110101, its octal number is _____, its hex number is _____.
6. $[22.2]_8 = [\quad]_2$,
 $[18.A]_{16} = [\quad]_2$,
 $[63]_{16} = [\quad]_8 = [\quad]_2$.
7. A decimal number 1102, its binary number is _____, its octal number is _____, its hex number is _____.
8. A binary number -0.1011, its 1s complement code is _____, its 2s complement code is _____.
9. A binary number -10010, its sign-magnitude code is _____, its 2s complement code is _____.
10. A binary number -0.001101, its 1s complement code is _____, its 2s complement code is _____.
11. An 8421 code number, 0101 0110, its decimal number is _____.
12. A 2421 code number, 1100 1110, its decimal number is _____.
13. When a digital system process analog signals, a/an _____ device is used to convert the signals.
14. The integer's sign-magnitude number has _____ (one/two) representation(s) of "0".
15. The integer's 2s complement number has _____ (one/two) representation(s) of "0".

B. Short Answer Questions:

1. How to convert a Gray code number to a binary number?

2. How to add error correction to a parity code system?

3. Convert the following binary number to the decimal, octal, and hex numbers, respectively.

(1) 1110101 (2) 0.1110101 (3) 10111.01

4. Convert the following decimal numbers to the binary, octal, and hex number, respectively.

(1) 29 (2) 0.27 (3) 33.33

5. Try to use 8421 code and Gray code to represent the following binary number:

(1) $(111110)_2$ (2) $(1100110)_2$